

Project Planning

PocketSmart AI – Your Smart Budget & Recommendation Assistant

1. Project Objective

The main objective is to develop an AI-powered budget recommendation platform that helps users plan lifestyle-related expenses and receive personalized recommendations.

2. Development Plan

The project is divided into the following stages:

1. Brainstorming
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 2. Requirement Analysis
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 3. Project Design
 - ↓
 4. Project Planning
 - ↓
 5. Development
 - ↓
 6. Testing
 - ↓
 7. Documentation
 - ↓
 8. Demonstration
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3. Planned Modules

Module 1 – Authentication

- Registration
- Login
- Logout
- JWT authentication
- Session management

Module 2 – Home Interior Planner

- Budget input
- Room selection
- Product requirements
- AI recommendations

Module 3 – Party Planner

- Budget input
- Guest count
- Event type
- Food
- Decoration
- Venue recommendations

Module 4 – Jewelry Planner

- Budget
- Occasion
- Style preference
- Outfit image
- AI-based matching

Module 5 – Recommendation History

- Store previous queries
- Display previous recommendations
- Reuse previous planning information

4. Technology Planning

Component	Technology
Programming Language	Python
Backend	FastAPI
AI Model	Gemini 1.5 Flash Pro
Frontend	HTML, CSS, JavaScript
Templates	Jinja2
Server	Uvicorn
Authentication	JWT
Communication	API / HTTP
Configuration	Environment Variables

5. Development Milestones

Milestone 1

Gemini AI initialization and API connectivity.

Milestone 2

Core planner functionality and recommendation logic.

Milestone 3

FastAPI backend integration.

Milestone 4

Frontend and user interface development.

Milestone 5

Testing and optimization.

6. Risk Planning

Potential challenges include:

- Invalid user inputs
- Insufficient AI recommendations
- API connectivity problems
- Incorrect budget interpretation
- Image-processing limitations
- External platform data availability

Planned Solutions

- Input validation
 - Prompt optimization
 - Fallback recommendations
 - API testing
 - Error handling
 - Modular architecture
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7. Expected Outcome

The completed system should provide users with a simple interface where they can enter their budget and requirements and receive AI-generated recommendations suitable for their selected planning category.